

CS 544: Formal Concept Analysis: Theory and Practice

Course Code	CS 544
Course Name	Formal Concept Analysis: Theory and Practice
L-T-P-C	3-0-0-3
Prerequisites	CS208 Mathematical Foundations of Computer Science or equivalent; IC250 Programming and Data Structure Practicum or equivalent
Intended for	B.Tech. (EE. & CSE) /MS/MTech/PhD
Distribution	Elective for B.Tech. III/IV year, MS, M.Tech., Ph.D.
Approval	23rd BoA (Apr 2018); 16th Senate (Apr 2018)

Course Contents

- **Formal contexts, formal concepts and concept lattice:** Formal context; conceptual scaling of many valued contexts; concept forming operators; basic mathematical structures behind FCA: Galois connections and closure operators; algebra of concepts - partial order and lattices; basic theorem; concept lattice diagrams. [8 Lectures]
- **Centralized and distributed algorithms for concept lattice construction:** Clarification and reduction of formal concepts; basic algorithms for computing concept lattice next closure algorithm, linear time closed item-set miner; distributed implementation of the algorithms for concept lattice construction naive approach, map-reduce based close-by-one, distributed closed item-set miner, trade-offs between breadth-first search and depth-first search based approaches; concept interestingness measures support, cue-validity, stability, lift and separation; incremental algorithms for constructing concept lattice. [12 Lectures]
- **The canonical basis:** Attribute implications, computing closure under implications, learning implications with membership and equivalence queries; pseudo-closed sets and canonical basis of implications, finding pseudo-closed sets; attribute exploration algorithm and its variations; concept exploration with partial lattice. [10 Lectures]
- **Applications of FCA:** Case-studies of concept exploration exploring faulty data, exploration in a fuzzy setting, triadic data; analysis of social networks social networks as formal contexts, individuality of social networks, knowledge communities; FCA in databases: learning functional dependencies, extensions to knowledge graphs; neural network architecture based on concept lattice. [12 Lectures]

Text Books

1. Bernhard-Ganter and Sergei Obeidkov, **Conceptual Exploration**, Springer 2016.
2. Rokia Missaoui, Sergei O. Kuznetsov and Sergei Obiedkov, **Formal Concept Analysis of Social Networks**, Lecture Notes in Social Networks, Springer 2017.

Reference Books:

1. Research papers on FCA
2. Open source softwares:
 - (a) <https://github.com/fcatools/>
 - (b) <http://www.upriss.org.uk/fca/fcasoftware.html>
3. Radim Belohlavek, **Introduction to Formal Concept Analysis**, Olomouc 2008.
4. Bernhard Ganter and Rudolf Wille, **Formal Concept Analysis: Mathematical Foundations**, Springer 1999.